

3.6 Biology Progress Check

HUMAN IMPACT ON THE ENVIRONMENT

IGC HK Exam



2. The Grand Banks is an area of sea off the coast of Newfoundland in Canada. It was once one of the most productive fishing grounds in the world for Atlantic cod.

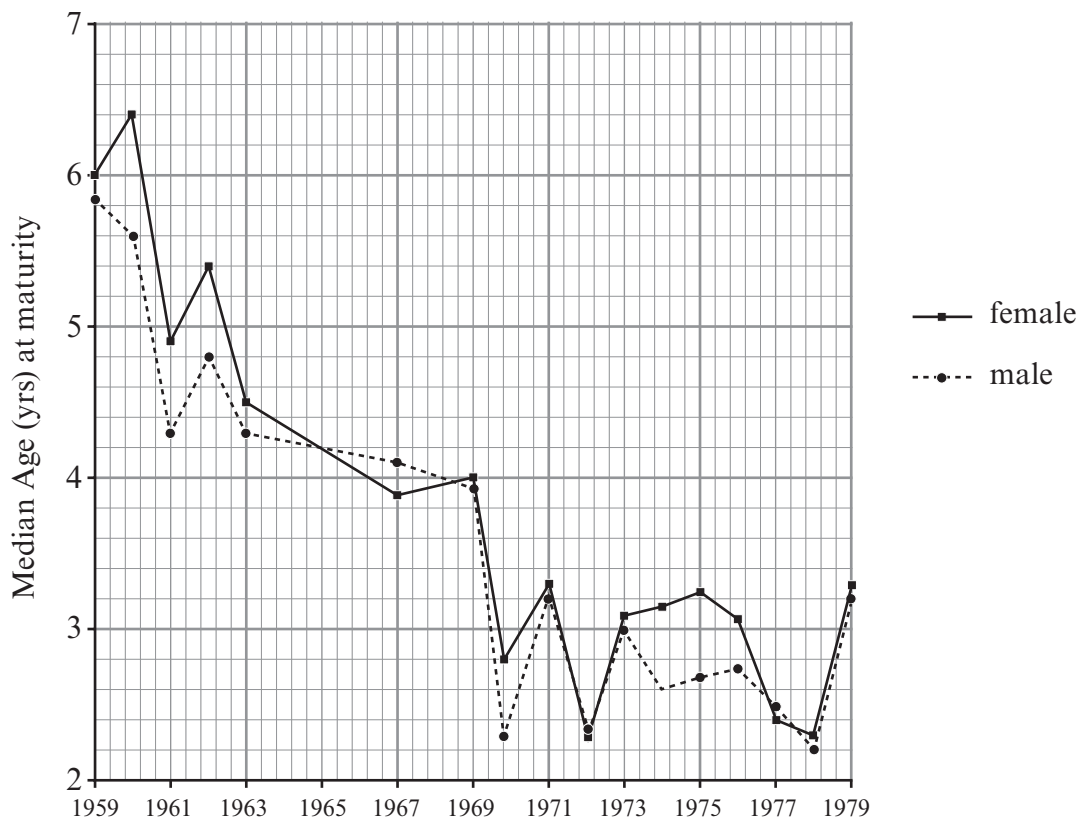
The cod was fished heavily for about 50 years.

About 60% of the total cod population of reproductive age was harvested annually.

Cod fishing in the Grand Banks was closed in 1992 but by then the population was less than 1% of what it had been.

Cod grow evenly throughout their life.

The cod that remained when fishing was finally closed were much smaller and grew more slowly than the cod that lived in the Grand Banks several decades previously.



Graph to show the median age of cod at sexual maturity in the same location during the time of heaviest fishing.

- (a) (i) Use the information provided opposite and your own knowledge of natural selection to describe and explain how the phenotype of the cod has changed since 1960. [5]

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- (ii) The cod fisheries have been closed for nearly 20 years but there has been little change in the phenotype and no population recovery. Suggest why there has been little change in the phenotype and no population recovery. [3]

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- (b) Other than restricting the mesh size of nets, give **two** other methods which are used to prevent overfishing. [2]

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- (c) (i) One solution to overfishing is aquaculture or fish farming.
Give **two** problems associated with producing fish in this way. [2]

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- (ii) Wild trout are diploid (**2n**). Some trout used in fish farming are triploid (**3n**).
Suggest why triploid trout are infertile. [4]

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Question			Marking details	Marks Available
2.	(a)	(i)	A. <u>Variation</u> in age at which sexual maturity is reached; B. Caused by mutation; C. Reach sexual maturity earlier/ Small fish {have a selective advantage/ pass through net}/ ora; D. Breed/ reproduce; <i>reject mate</i> E. Pass on alleles to offspring; <i>reject genes</i> F. Allele frequency for earlier maturity / hence small size at maturity increases; G. Figs quoted from graph (in context);	Max 5
		(ii)	Very few large cod survived/ ORA; <i>reject none</i> reduced gene pool; {No/ little} mutation (to increase size) / insufficient time for genetic drift (to increase size) / No gene flow from another gene pool; Small fish produce less gametes/ difficulty in breeding/ few fish remain to reproduce/ reproductive isolation; Not enough food/ increased competition for food/ increased predation/ disease; Change in {temperature/ pH}/ pollution;	Max 3
	(b)		Restricted fishing times/ hours; Quotas/ licenses; Exclusion zones/ OWTTE; Limiting numbers of fishing vessels/ international agreements limiting catches; Limiting season; Restriction of <u>area</u> of nets; Closing spawning and/ or nursery areas; <i>REJECT any reference to mesh size</i>	2

Question			Marking details	Marks Available
	(c)	(i)	Eutrophication/ pollution; {Disease/ parasites} more likely (to spread) in {cultivated fish/ overcrowded conditions}/ disease may spread to wild fish; {Antibiotics/ pesticides} qualified e.g. can harm other marine organisms/ bioaccumulation of pesticides/ enters food chain/ high cost; Problems associated with flow of alleles into wild population; Higher level of dioxins/ PCBs in farmed fish;	2
		(ii)	Three of each type of chromosome / {odd/uneven} number of chromosomes/ unpaired chromosomes; No pairing of <u>homologous</u> chromosomes/ no bivalent formed; Prophase 1 meiosis; Meiosis does not take place; No gametes produced; Question 2 total	Max 4 [16]